

Design a Hebb Network for implementing Logical AND gate considering bipolar inputs and target.

Hebbian Learning Rule

Inputs				Weight Changes			New Weights		
x1	x2	b	y	$\Delta w1$	$\Delta w2$	Δb	w1(new)	w2(new)	b(new)
				x1 y	x2 y	y	w1(old)+ $\Delta w1$	w2(old)+ $\Delta w2$	b(old)+ y
1	1	1	1						
1	-1	1	-1						
-1	1	1	-1						
-1	-1	1	-1						

Inputs				Weight Changes			New Weights		
x1	x2	b	y	$\Delta w1$	$\Delta w2$	Δb	w1(new)	w2(new)	b(new)
				x1 y	x2 y	y	w1(old)+ $\Delta w1$	w2(old)+ $\Delta w2$	b(old)+ y
1	1	1	1	= 1 * 1 = 1	= 1 * 1 = 1	= 1	0+1 = 1	0+1 = 1	0+1 = 1
1	-1	1	-1						
-1	1	1	-1						
-1	-1	1	-1						

Inputs				Weight Changes			New Weights		
x1	x2	b	y	$\Delta w1$	$\Delta w2$	Δb	w1(new)	w2(new)	b(new)
				x1 y	x2 y	y	w1(old)+ $\Delta w1$	w2(old)+ $\Delta w2$	b(old)+ y
1	1	1	1	1	1	1	$0+1$ = 1	$0+1$ = 1	$0+1$ = 1
1	-1	1	-1	-1	1	-1	$1+(-1)$ = 0	$1+1$ = 2	$1+(-1)$ = 0
-1	1	1	-1						
-1	-1	1	-1						

Inputs				Weight Changes			New Weights		
x1	x2	b	y	$\Delta w1$	$\Delta w2$	Δb	w1(new)	w2(new)	b(new)
				x1 y	x2 y	y	w1(old)+ $\Delta w1$	w2(old)+ $\Delta w1$	b(old)+ y
1	1	1	1	1	1	1	0+1 = 1	0+1 = 1	0+1 = 1
1	-1	1	-1	-1	1	-1	1+(-1) = 0	1+1 = 2	1+(-1) = 0
-1	1	1	-1	1	-1	-1	0+1 = 1	2+(-1) = 1	0+(-1) = -1
-1	-1	1	-1	1	1	-1	1+1 = 2	1+1 = 2	-1+(-1) = -2